FINITE TYPENESS FOR A MORPHISM OF COUNTABLY INDEXED AFFINE FORMAL ALGEBRAIC SPACES AND THE CORRESPONDING RING PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 7 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Affineness and representability of a morphism of affine formal algebraic spaces. Same chapter, Lemma lemma-affine-representable-by-algebraic-spaces, invoked at proof lines 6709-6711. Makes the morphism affine once it is locally of finite type.
- The type ascends along a representable affine morphism over an affine formal algebraic space. Same chapter, Lemma lemma-property-goes-up-affine-morphism, invoked at proof lines 6711-6713. Gives that the source is countably indexed.
- Countably indexed affine formal algebraic spaces as formal spectra, and morphisms between them as continuous ring maps. Same chapter, Lemma lemma-countably-indexed at proof lines 6715-6717 and Lemma lemma-morphism-between-formal-spectra at proof lines 6717-6719. Convert the geometric statement into a statement about a ring map.
- The comparison between finite typeness of the morphism and tautness plus topological finite typeness of the ring map, and the finite typeness conditions of the earlier lemma. Same chapter, Lemma lemma-topologically-finite-type-finite-type at proof lines 6719-6721 and Lemma lemma-quotient-restricted-power-series referred to in the statement. Give the final equivalence.

Source reference. Original lemma 0AKN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4209–4209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0AN1: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2493–2493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN0: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2355–2355. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKP: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4316–4316. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0ANU: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 6560–6560. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0ANT: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 6343–6343. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 0ANA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4953–4953. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of affine formal algebraic spaces. Assume Y countably indexed. The following are equivalent*

(1) *f is locally of finite type,*
(2) *f is of finite type,*
(3) *f corresponds to a morphism $B \to A$ of $WAdm^{count}$ (Section 0ANA) satisfying the equivalent conditions of Lemma 0ANU.*

Proof. Since X and Y are affine it is clear that conditions (1) and (2) are equivalent. In cases (1) and (2) the morphism f is representable by algebraic spaces by definition, hence affine by Lemma 0AKN. Thus if (1) or (2) holds we see that X is countably indexed by Lemma 0AKP. Write $X = \mathrm{Spf}(A)$ and $Y = \mathrm{Spf}(B)$ for topological S-algebras A and B in $WAdm^{count}$, see Lemma 0AN1. By Lemma 0AN0 we see that f corresponds to a continuous map $B \to A$. Hence now the result follows from Lemma 0ANT. □